CMPE 49G – Fundamentals of Particle-based Simulations

Homework 2

Chat with LLM on Particle Image Velocimetry (PIV)

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LLM Used: *Claude Sonnet 4.6*

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1 AI Transparency Note

Disclosure: This report documents a structured chat session with **Claude Sonnet 4.6** on the topic of Particle Image Velocimetry (PIV). The questions were formulated by the student following the homework outline. The usage level is **heavy**: the LLM was the primary source of information, and answers were reviewed and verified against standard PIV literature (Raffel et al., *Particle Image Velocimetry: A Practical Guide*, 3rd ed.). The student takes full responsibility for the submitted content.

2 Chat Log

2.1 Q1: What is PIV? What is it used for in real life?

Student Question

What is Particle Image Velocimetry (PIV)? What is it used for in real life?

LLM Answer

Particle Image Velocimetry (PIV) is an **optical, non-intrusive flow measurement technique** used to measure the velocity field of a fluid (liquid or gas) over a 2D plane—or a 3D volume in stereoscopic/tomographic variants—at an instant in time.

How It Works Conceptually

Small tracer particles chosen to closely follow the fluid motion (*neutrally buoyant*) are seeded into the flow. A laser sheet illuminates a thin plane of the flow at two closely spaced instants, t and $t + \Delta t$. A camera captures two frames. By statistically comparing how particle groups have moved between the two frames, the local velocity vector field is computed across the entire measurement plane.

Real-life Applications:

Domain	Example
Aerospace	Wind tunnel studies of airfoil wakes, turbulent boundary layers
Automotive	Flow around vehicles, engine combustion studies
Biomedical	Blood flow in arteries (micro-PIV), heart valve hemodynamics
Civil/Environmental	River flow, coastal wave dynamics
Turbomachinery	Pump and turbine internal flows
Fundamental Research	Turbulence, vortex dynamics, mixing layers

The key advantage over intrusive probes (hot-wire anemometry, Pitot tubes) is that PIV captures the **whole-field velocity** simultaneously without disturbing the flow.

PIV Velocity Field

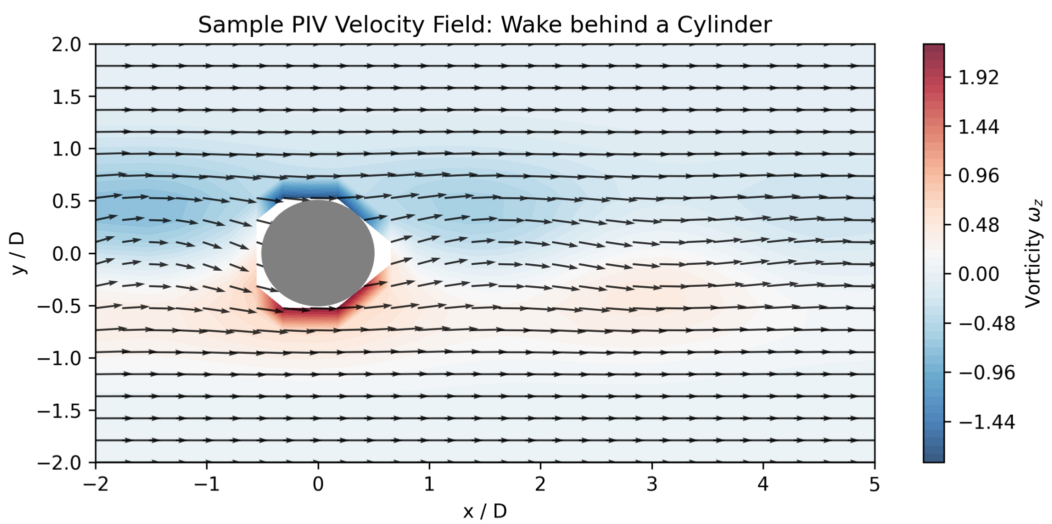

Figure 1: Sample PIV velocity vector field overlaid on vorticity contours (simulated wake behind a cylinder).

2.2 Q2: What does "non-intrusive" mean?

Student Question

You mentioned PIV is "non-intrusive." What does that exactly mean?

LLM Answer

"Non-intrusive" means the measurement technique does not physically insert any sensor or probe into the fluid. A Pitot tube or hot-wire anemometer must be placed inside the flow, which:

- disturbs the flow field around the probe,
- measures only at a single point,
- can be impractical in hostile environments (high-temperature combustion, biological tissues).

PIV uses light (laser) to illuminate and a camera to observe—both from outside the flow domain. The only things inside the fluid are tracer particles, which are small enough (typically 1–100 µm) that they faithfully follow fluid streamlines without altering the flow behavior.

2.3 Q3: What are the main physical components of a PIV system?

Student Question

What are the main physical components required for a PIV system?

LLM Answer

1. Tracer Particles (Seeding)

Small particles that **follow the flow** (low Stokes number, $St \ll 1$). Common choices: hollow glass spheres (liquids), oil droplets, titanium dioxide (TiO_2) powder (gas flows), polystyrene microspheres (micro-PIV). Particle diameter typically 1–100 µm.

The Stokes number is defined as:

$$St = \frac{\tau_p}{\tau_f} \tag{1}$$

where τ_p is the particle response time and τ_f is the characteristic flow time scale. For good flow tracking, $St \ll 1$.

2. Illumination System (Pulsed Laser)

A **double-pulsed Nd:YAG laser** is most common (wavelength 532 nm, green). Two pulses are fired with a precise inter-frame time Δt. A set of cylindrical and spherical lenses shapes the beam into a thin **laser sheet** (typically 1–2 mm thick).

3. Camera

A **CCD or sCMOS camera** with double-frame (frame-straddling) capability to capture two images synchronized with the laser pulses. Resolution typically 1–16 Megapixels. The camera is placed perpendicular to the laser sheet for standard 2D PIV.

4. Synchronizer (Timing Unit)

A **hardware timing controller** that precisely synchronizes the laser pulses with the camera exposure. Controls Δt, which determines the measurable velocity range.

5. Optical Access / Flow Facility
The experimental setup must have **optical access**—transparent walls (glass or acrylic) for the laser sheet and camera view.

6. Computer + PIV Software
Acquires image pairs and processes them to extract velocity vectors using cross-correlation algorithms.

2.4 Q4: What is the mathematics behind PIV?

Student Question

What is the mathematics behind PIV? How are velocity vectors computed from images?

LLM Answer

The core mathematical engine of PIV is **cross-correlation**.

Step 1 – Image Acquisition
Two images $I_1(\mathbf{x})$ and $I_2(\mathbf{x})$ are recorded at times t and $t + \Delta t$, where $\mathbf{x} = (x, y)$ is the pixel coordinate.

Step 2 – Interrogation Window Division
Each image is divided into small overlapping rectangular regions called **interrogation windows** (IW), typically 16×16 to 64×64 pixels.

Step 3 – Cross-Correlation
For each interrogation window, the displacement of particles is found via:

$$C(\mathbf{s}) = \sum_{\mathbf{x} \in W} I_1(\mathbf{x}) \cdot I_2(\mathbf{x} + \mathbf{s}) \tag{2}$$

where $\mathbf{s} = (s_x, s_y)$ is the displacement vector and W is the interrogation window domain. The displacement that maximizes C is the most probable particle displacement:

$$\mathbf{d} = \arg\max_{\mathbf{s}} C(\mathbf{s}) \tag{3}$$

Step 4 – FFT-based Acceleration
Direct computation is $O(N^4)$. Using the **cross-correlation theorem** with the Fast Fourier Transform (FFT):

$$C(\mathbf{s}) = \mathcal{F}^{-1}\left\{\hat{I}_1(\mathbf{k}) \cdot \hat{I}_2^*(\mathbf{k})\right\} \tag{4}$$

where $\hat{I} = \mathcal{F}\{I\}$ and * denotes the complex conjugate. This reduces complexity to $O(N^2 \log N)$.

Step 5 – Sub-pixel Peak Detection Sub-pixel accuracy (0.05–0.1 pixel) is achieved by fitting a **Gaussian curve** to the correlation peak:

$$x_{\text{peak}} = i + \frac{\ln C(i-1) - \ln C(i+1)}{2\ln C(i-1) - 4\ln C(i) + 2\ln C(i+1)} \tag{5}$$

Cross-Correlation Peak

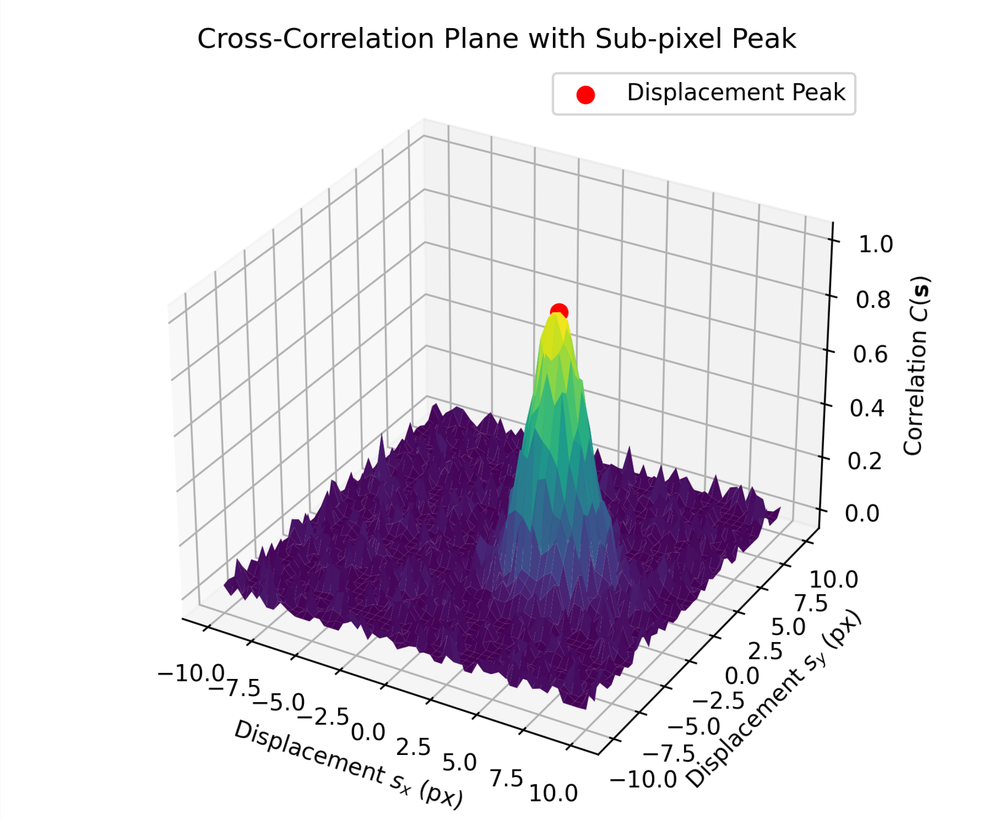

Figure 2: 3D representation of the cross-correlation plane $C(\mathbf{s})$ showing the sub-pixel correlation peak that identifies the most probable displacement vector.

Step 6 – Velocity Calculation
Physical velocity is recovered as:

$$\mathbf{u} = \frac{\mathbf{d} \cdot M_{\text{scale}}}{\Delta t} \tag{6}$$

where M_{scale} (mm/pixel) is the image magnification obtained from calibration.

Step 7 – Vector Validation

Spurious vectors (outliers) are detected using:

- **Normalized Median Test (NMT):** A vector is flagged if it deviates from the median of its neighbors by more than a threshold.
- **Signal-to-Noise Ratio (SNR):** Ratio of primary to secondary correlation peak.

Outliers are replaced by interpolation from neighboring valid vectors.

2.5 Q5: What is a PIV software?

Student Question

What is a PIV software? Are there well-known examples?

LLM Answer

A **PIV software** is a program that automates the complete processing pipeline: reading raw image pairs, running cross-correlation algorithms, validating vectors, and visualizing the resulting velocity field.

Software	Type	Notes
DaVis (LaVision)	Commercial	Industry standard, very feature-rich
PIVlab	Free / MATLAB	Most widely used in academia, open-source
OpenPIV	Free / Python	Open-source, Python + GPU support
INSIGHT 4G (TSI)	Commercial	Used with TSI hardware systems
fluidimage / mpivpy	Free / Python	Research-oriented, Python ecosystem

PIVlab is especially popular because it is open-source, runs in MATLAB, and implements state-of-the-art algorithms with a graphical user interface.

2.6 Q6: What are the most important modules for a PIV software?

Student Question

What are the most important modules that a PIV software must have?

LLM Answer

1. Image Pre-processing Module
Improves image quality before correlation: background subtraction, intensity normalization, CLAHE (Contrast Limited Adaptive Histogram Equalization), and masking of solid walls.

2. Interrogation / Cross-Correlation Engine
The core computational module: FFT-based cross-correlation, multi-pass iterative window refinement, and ensemble correlation for low-seeding-density flows.

3. Vector Validation Module
Detects and replaces outliers using the Normalized Median Test, global SNR filtering, and iterative interpolation.

4. Post-processing Module
Derives secondary quantities from the velocity field:

$$\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \quad \text{(vorticity)} \tag{7}$$

Also: divergence, strain rate, turbulent kinetic energy, Reynolds stresses.

5. Calibration Module
Converts pixel displacements to physical units using a reference calibration target (dot grid or checkerboard). Also corrects lens distortion.

6. Visualization Module
Vector field plots, color-contoured scalar maps, streamline plots, and time-series animations.

7. I/O Module
Reads raw camera images (TIFF, BMP, HDF5) and exports results (ASCII, HDF5, VTK) for further CFD post-processing.

2.7 Q7: What are the mathematical techniques used in PIV software?

Student Question

What are the specific mathematical techniques implemented inside PIV software?

LLM Answer

1. Window Deformation (Image Deformation PIV)
The interrogation window in I_2 is **deformed** according to the local velocity gradient using bilinear or B-spline interpolation:

$$I_2^{\mathrm{def}}(\mathbf{x}) = I_2(\mathbf{x} + \mathbf{u}(\mathbf{x}) \cdot \Delta t) \tag{8}$$

This significantly improves accuracy near shear layers.

2. Multi-Resolution / Multi-Pass Algorithm

A pyramid approach: Pass 1 uses large IW (64×64, coarse), Pass 2 uses 32×32, Pass 3 uses 16×16 with the previous result as the initial offset. This reduces peak-locking artifacts.

3. Sub-pixel Interpolation Schemes

Gaussian 3-point fit (most common), parabolic fit, or centroid method – all achieve 0.05–0.1 pixel accuracy.

4. Spatial Differentiation

Velocity gradients via central finite differences:

$$\frac{\partial u}{\partial x} \approx \frac{u_{i+1,j} - u_{i-1,j}}{2\Delta x} \tag{9}$$

5. Reynolds Decomposition and Turbulence Statistics

For N instantaneous fields:

$$u(\mathbf{x}, t_n) = \bar{u}(\mathbf{x}) + u'(\mathbf{x}, t_n), \qquad \bar{u} = \frac{1}{N}\sum_{n=1}^{N} u(\mathbf{x}, t_n) \tag{10}$$

Turbulent kinetic energy:

$$k = \frac{1}{2}\left(\overline{u'^2} + \overline{v'^2}\right) \tag{11}$$

6. Proper Orthogonal Decomposition (POD)

Applied to time-series PIV data to extract dominant flow structures ordered by energy content (snapshot method):

$$\mathbf{u}(\mathbf{x}, t) = \sum_{i=1}^{N} a_i(t)\, \boldsymbol{\phi}_i(\mathbf{x}) \tag{12}$$

7. Spectral Analysis

Spatial power spectral density via 2D FFT of velocity fluctuation fields; used to verify turbulent energy cascades following Kolmogorov's $-5/3$ law.

8. Optical Flow and Deep Learning Methods

Modern PIV software optionally uses optical flow algorithms (Horn–Schunck, Lucas–Kanade) or CNN-based approaches (e.g., LiteFlowNet adapted for PIV), achieving higher spatial resolution than correlation-based methods.

2.8 Q8: What is the Stokes number in more detail?

Student Question

You mentioned the Stokes number earlier. Can you explain it in more detail?

LLM Answer

The **Stokes number** (St) quantifies how well a particle follows the fluid flow:

$$St = \frac{\tau_p}{\tau_f} \tag{13}$$

Particle relaxation time for a spherical particle in the Stokes drag regime:

$$\tau_p = \frac{\rho_p\, d_p^2}{18\, \mu_f} \tag{14}$$

where ρ_p is the particle density, d_p is the particle diameter, and μ_f is the dynamic viscosity of the fluid.

Interpretation:
- $St \ll 1$: Particle follows the fluid perfectly — ideal PIV tracer.
- $St \approx 1$: Particle starts to deviate from the fluid streamlines.
- $St \gg 1$: Particle motion is dominated by inertia, independent of the flow.

In practice, PIV particles are designed to satisfy $St < 0.1$.

2.9 Q9: What are the limitations of PIV?

Student Question

What are the limitations and weak points of PIV as a technique?

LLM Answer

Limitations of PIV

1. **Optical access required:** The flow domain must have transparent walls. Impossible in many industrial settings (metal pipes, engines).
2. **2D projection problem:** Standard 2D PIV measures only two velocity components. The out-of-plane component (w) is lost. Stereoscopic PIV (3C-2D) or Tomographic PIV (3D) address this but at higher cost.
3. **Seeding quality is critical:** Non-uniform seeding creates void regions; over-seeding causes particle overlap. Maintaining $\sim$5–10 particle pairs per IW is practically challenging.
4. **Spatial resolution limit:** Features smaller than the interrogation window are spatially filtered out.
5. **High-speed flows:** Very fast flows require sub-microsecond Δt and high-frame-rate cameras — expensive hardware.
6. **Near-wall measurements:** Laser reflection from solid walls creates a glare region that obscures particle images near the wall ($y^+ < 5$), making accurate boundary layer measurement difficult.
7. **Temporal resolution:** Standard double-frame PIV yields one velocity field per laser shot pair. Time-resolved PIV (TR-PIV) requires high-repetition-rate lasers and high-speed cameras.

2.10 Q10: Raw Follow-up on Near-Wall Limitations

Student (Raw Prompt): "You said laser reflection makes near-wall measurements hard. Is there any trick to fix that or are we just blind near the wall?"

LLM Answer:

We are not entirely blind, but it requires special tricks! The most common workarounds are:

- **Fluorescent particles:** Using Rhodamine-B dyed particles and placing an optical filter in front of the camera. The filter blocks the glaring green laser light (532 nm) but lets the orange fluorescent emitted light (around 580 nm) pass through.
- **Painting the wall:** Coating the background/wall with black matte paint (or Rhodamine paint) to absorb the laser sheet and reduce reflections.
- **Index matching:** Matching the refractive index of the fluid with the solid wall material if possible.

3 Summary of Lessons Learned

Key Takeaways

- PIV is a non-intrusive whole-field optical velocimetry technique based on tracking particle displacements between two laser-illuminated frames.
- The core mathematics relies on FFT-accelerated cross-correlation of particle image pairs, with sub-pixel Gaussian peak fitting for accuracy.
- The Stokes number ($St \ll 1$) is the fundamental criterion for choosing tracer particles that faithfully follow the flow.
- Key hardware components are a double-pulsed Nd:YAG laser, a CCD/sCMOS double-frame camera, a hardware synchronizer, and seeding particles.
- Advanced software modules implement window deformation, multi-pass iterative refinement, POD decomposition, and even deep learning-based optical flow.
- 2D PIV only measures two out of three velocity components — a fundamental physical limitation addressed by stereoscopic and tomographic variants.
- Near-wall measurements and optical access requirements are the most practically limiting constraints of PIV.

Personal Reflection I tried running PIVlab on a sample dataset (flow past a cylinder) during this homework to see these concepts in action. I noticed that near the boundaries of the cylinder, the velocity vectors were often erroneous or chaotic unless I specifically drew a mask over the solid body. It perfectly demonstrated the "near-wall measurements" limitation discussed with the LLM. Furthermore, tweaking the interrogation window size down to 16×16 drastically improved the resolution of the small turbulent eddies in the wake, though it increased the processing time significantly.

4 Discussion on Weak Points of the LLM

Critical Observations

1. **Omission of Shake-The-Box (STB):** The LLM did not spontaneously mention *Shake-The-Box*, a state-of-the-art Lagrangian particle tracking method that outperforms classical tomographic PIV in 3D flows. This is a significant modern development that a complete answer should have included.
2. **Deep learning claims:** The LLM cited LiteFlowNet adapted for PIV as a deep learning technique. While conceptually correct, the specific network name should be verified against the latest literature (e.g., PIV-LiteFlowNet-en by Cai et al., 2019), as the field evolves rapidly. The LLM did not provide bibliographic references.
3. **No mention of Micro-PIV:** Micro-PIV for microscale biological flows was only briefly mentioned in passing (Q1 table) but not discussed as a distinct variant with its own challenges (evanescent wave illumination, Brownian motion noise, depth-of-field effects).
4. **Ensemble correlation not elaborated:** The LLM listed ensemble correlation as a module feature but did not explain it in detail. Ensemble/sum-of-correlation

is a critical technique for low-seeding-density flows and deserved more explanation.

5. **No uncertainty quantification discussion:** Modern PIV software includes uncertainty quantification (UQ) methods (e.g., peak-ratio methods, correlation statistics). The LLM did not mention this aspect at all.

These gaps do not indicate incorrect answers but rather incomplete coverage. The student verified the provided information against Raffel et al. ***Particle Image Velocimetry: A Practical Guide*** *(3rd ed., Springer, 2018) and found the core content to be accurate.*